1. Introduction

- Communications Tasks
- Communications model

2. Signal Analysis

- Signalling concepts
- Time/Frequency domain representations
- Bandwidth concepts – Transmission Signal Bandwidth and Transmission System Bandwidth
- Relationship between Data Rate and Bandwidth

3. Data Transmission concepts

- Digital versus analogue data/signals/systems

4. Transmission Impairments

- Attenuation
- Noise

5. Channel capacity

- Nyquist
- Shannon

6. Transmission Media

- Wired – TP/Co-ax/OF
- Wireless – Microwave/Radio/Satellite
- Full Duplex Communications

7. Data Encoding

- Digital data onto Digital signals – NRZ, Diff. Manchester etc.
- Digital data onto Analogue signals – FSK, PSK etc.
- Analogue data onto Digital signals – PCM, Delta Modulation.
- Analogue data onto Analogue signals – AM, FM etc.

8. Synchronous Transmission

- Timing problems
- Framing

9. Data Link Control

- Basic Requirements

(a) Flow Control techniques

- Stop and Wait – single frame transmission
- Sliding Window – use of extended memory buffers
- Link Utilization

(b) Error detection techniques

- Parity/Checksums/CRC – operation and examples

(c) Error Control techniques

- Stop and Wait/Go-back-N/Selective Reject ARQ – operation and examples

(d) Sample protocol – HDLC

- Frame format – fields, bit stuffing etc.
- Commands/Responses
- Operation for various scenarios such as busy, missing frame etc.
- Status of incoming/outgoing buffers for data and response frames arriving and departing

10. Multiplexing

- FDM and STDM - components and operation
- Carrier Systems

11. Protocol Architecture

- ISO-OSI 7-Layer model

12. Local Area Networks

- LAN applications – PC/Backend/Backbone
- LAN Topologies and operation – Bus/Ring/Star
- Medium Access Control techniques including Collision Detection and Collision Avoidance – Bus, Wireless and Ring LANs
- Example LAN network technologies – Ethernet, IBM Token Ring, etc.
- Extending LANs using Repeaters and Bridges
- MAC/Hardware Addressing
- MAC frame formats
- Bridges – operation, functionality, routing (frame filtering) and routing table

13. Internetworks

- Universal service provision and associated issues
- Routers and protocol software
- IP Addressing (classful, classless, netid/hostid etc.)
- IP Address Allocation and Address Tables
- IP Datagram Format
- Forwarding IP datagrams, routing tables, fragmentation etc.
- Encapsulation
- ARP,
- NAT.